



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO ARUNACHALAM FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

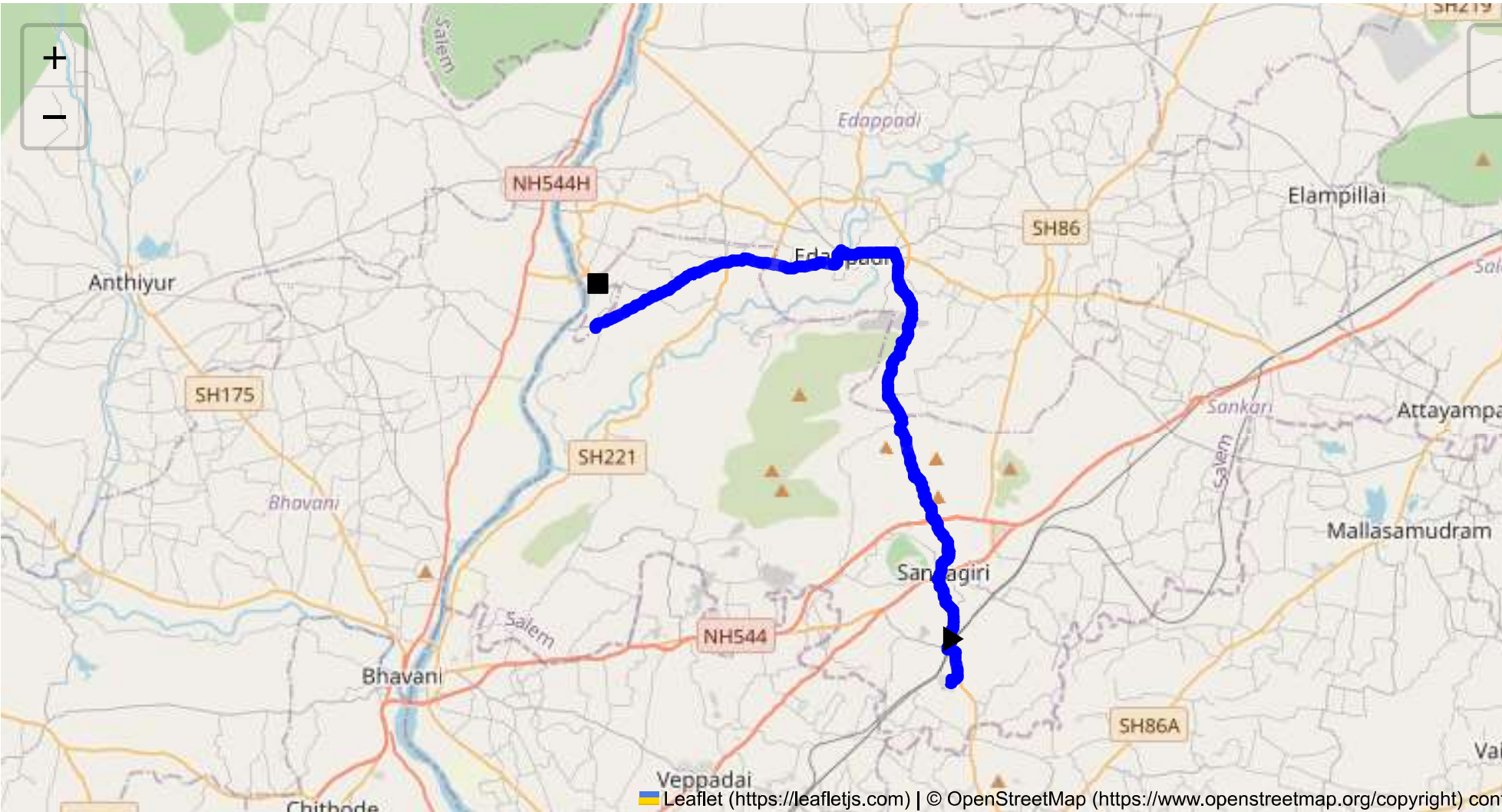
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 30.53 km**  
**Estimated Duration: 0.8 hours**  
**Adjusted Duration (Heavy Vehicle): 1.0 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.559318, 77.749811)**



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

**Overview of the Route Map:** The route from Sangagiri (CVQF+23W) to Kottayur (HP5X+MXQ) covers approximately 30.53 kilometers through Tamil Nadu. The path includes several state highways and rural roads. The terrain is mainly flat with some areas of farmland and small villages.



**Typical Weather Conditions and Potential Weather-related Hazards:** Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The monsoon season from October to December can bring heavy rains and potential flooding, which may cause visibility issues and slippery roads. Summers are hot and dry, while the winter season is mild, making driving conditions generally favorable outside the monsoon period.

**Analysis of Traffic Patterns:** Traffic congestion in this region is primarily affected by local market activity and school timings. Peak hours generally occur between 7-10 AM and 5-8 PM. Sangagiri, being closer to the NH44, experiences moderate traffic, while rural and village roads may have lesser congestion but occasional local vehicular challenges like agricultural vehicles.

**Assessment of Road Quality and Infrastructure:** The roads within this stretch vary in quality. State highways are generally well-maintained, but smaller rural roads may have potholes and narrower lanes. Signage is present but may sometimes be lacking in smaller villages. There could be occasional construction work, impacting travel time.

**Suggestions for Alternative Routes:** In case of emergencies, alternative routes via NH44 could be considered, albeit with a slightly longer distance. Local roads connecting through adjacent villages may be leveraged to circumvent congested areas or obstacles.

**Summary of Local Regulations Affecting Hazardous Material Transport:** Transport of hazardous materials must comply with Indian Motor Vehicles Act regulations concerning load limits and safety measures. It is essential to ensure proper labeling and documentation of hazardous materials, with certain restrictions on time-of-day travel in urban areas to mitigate risk.

**Overview of Historical Incidents:** While this specific route does not have a high record of incidents involving heavy vehicles or hazardous material, nearby major highways, like NH44, have seen accidents primarily due to speeding or adverse weather conditions. Local knowledge and caution are recommended.

**Environmental Considerations and Sensitive Areas:** The route passes through agricultural areas and near some water bodies, requiring caution to avoid contamination from hazardous materials. Efforts should be made to minimize environmental impact by adhering to designated rest areas and avoiding unauthorized stops near sensitive zones.

**Analysis of Communication Coverage:** The communication network is generally stable near urban centers like Sangagiri, though certain rural sections between villages might experience weak signals or dead zones. It is advisable to have offline navigation tools as a backup.

**Estimated Emergency Response Times:** Response times may vary depending on the location. Urban areas may expect quicker responses, typically 20-30 minutes, whereas rural areas might experience longer delays, stretching to about an hour due to distance and road conditions.

**Overall Summary of Risk Assessment:** The route poses a moderate risk due to factors like weather conditions during the monsoon, road quality, and partial communication dead zones. Careful attention to truck maintenance, strict adherence to safety protocols, especially during peak traffic hours, and weather monitoring is advised to ensure safe transit. Preparing for rapid response and having clear communication strategies are essential parts of risk management for this route.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.58435, 77.83651 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2  | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5  | Turn       | High       | 11.45007, 77.87201 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.47407, 77.86839 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.47451, 77.86914 | 15 KM/Hr    |
| 8  | Turn       | High       | 11.47920, 77.86989 | 15 KM/Hr    |
| 9  | Turn       | Medium     | 11.48058, 77.87163 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.48155, 77.87174 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.48277, 77.87250 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.49397, 77.86740 | 30 KM/Hr    |
| 13 | Turn       | High       | 11.58533, 77.83450 | 15 KM/Hr    |
| 14 | Turn       | High       | 11.56019, 77.74944 | 15 KM/Hr    |

Emergency Locations

|   | type     | name                             | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------------|------------------------|-------------|------------|
| 0 | police   | Police Quarters                  | 11.4741243, 77.8673495 | 30 km/h     | Medium     |
| 1 | hospital | Kathir Eye Hospital              | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 2 | hospital | Thilagam Hospital                | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 3 | clinic   | Meyyazhagan Clinic               | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 4 | hospital | Dr. Jaganathan Hospital          | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 7 | hospital | Sankari Hospital                 | 11.4777496, 77.873892  | 30 km/h     | Medium     |
| 8 | hospital | KRS Hospital and Diabetic Centre | 11.5848828, 77.8449664 | 30 km/h     | Medium     |
| 9 | hospital | Kannan Hospital                  | 11.5854802, 77.8425936 | 30 km/h     | Medium     |

|    | type     | name                          | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------|------------------------|-------------|------------|
| 10 | hospital | Aravind Hospital              | 11.5854615, 77.8424165 | 30 km/h     | Medium     |
| 11 | clinic   | Uma Clinic                    | 11.58398, 77.83977     | 30 km/h     | Medium     |
| 12 | hospital | Muthu Eyecare Hospital        | 11.584736, 77.8393794  | 30 km/h     | Medium     |
| 13 | clinic   | KGP Clinic                    | 11.5847763, 77.8387488 | 30 km/h     | Medium     |
| 14 | hospital | Government Hospital, Edapadi  | 11.5848264, 77.8380174 | 30 km/h     | Medium     |
| 15 | hospital | Sri Sankar Hospital           | 11.5862984, 77.8355021 | 30 km/h     | Medium     |
| 16 | hospital | Sri Advika Hospital           | 11.585049, 77.834422   | 30 km/h     | Medium     |
| 17 | clinic   | Raj Physiotherapy Clinic      | 11.5857357, 77.8341218 | 30 km/h     | Medium     |
| 18 | hospital | Dr. O.S. Subramanium Hospital | 11.584996, 77.834235   | 30 km/h     | Medium     |
| 19 | hospital | Aravinthu Eye Hospital        | 11.5866631, 77.8341465 | 30 km/h     | Medium     |
| 20 | clinic   | Xcellent Homecare Clinic      | 11.584762, 77.8338085  | 30 km/h     | Medium     |
| 21 | hospital | Mohan Hospital                | 11.5841331, 77.8336277 | 30 km/h     | Medium     |
| 22 | hospital | Nivetha Hospital              | 11.5858509, 77.833451  | 30 km/h     | Medium     |
| 23 | hospital | Rathanasamy Hospital          | 11.58583, 77.833443    | 30 km/h     | Medium     |
| 24 | hospital | SKM Hospital                  | 11.5841611, 77.8334397 | 30 km/h     | Medium     |
| 25 | clinic   | Krishna Hospital and Clinic   | 11.5846542, 77.8333239 | 30 km/h     | Medium     |
| 26 | hospital | Arunagiri Hospital            | 11.5837095, 77.8332129 | 30 km/h     | Medium     |
| 27 | hospital | Pachiappa Hospital            | 11.58571, 77.833131    | 30 km/h     | Medium     |
| 28 | clinic   | Rakshan Polyclinic            | 11.5809367, 77.8326223 | 30 km/h     | Medium     |
| 29 | hospital | Siddha Hospital               | 11.5827668, 77.8022539 | 30 km/h     | Medium     |

Crowded Spots

|   | type        | name                                   | coordinates            | speed_limit | risk_level |
|---|-------------|----------------------------------------|------------------------|-------------|------------|
| 5 | school      | Goverment Boys Higher Secondary School | 11.477659, 77.8659632  | 30 km/h     | Medium     |
| 6 | marketplace | Sunday Market                          | 11.4796502, 77.8708458 | 30 km/h     | Medium     |

Route Photos of Risky Spots





**Risk Type: Roundabout**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.58435, 77.83651**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44029, 77.87544**





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47407, 77.86839



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47451, 77.86914





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.47920, 77.86989



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.48058, 77.87163





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.48155, 77.87174



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.48277, 77.87250



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49397, 77.86740





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58533, 77.83450



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56019, 77.74944

